

DECONSTRUCTING THE SOFTWARE FACTORY:

A Practical Application of Interorganizational Network Analysis



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Over the past 5 years, the number of DoD software organizations that employ nontraditional organizational structures has increased. These organizations, commonly referred to as software factories, often employ the network-based organizational structures found within high-technology industries. This article details ways in which network analysis techniques can be used to create a big picture view of these nontraditional organizations. Drawing on methodologies employed by network researchers, the authors develop and present an interorganizational analysis process that highlights a program's social and economic structures. Following the case history approach, they demonstrate the applicability of this approach by analyzing an emergent DoD software factory. Throughout the article, the authors discuss how network principles and accessible analysis techniques can be applied to real-world challenges faced by modern, network-based DoD organizations. Finally, they conclude by presenting the results of their analysis and a framework that the acquisition community can use to analyze nontraditional programs.

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Program managers (PMs) have various assessment and tracking tools at their disposal designed to provide actionable insights into the cost, schedule, and performance of acquisition programs. Existing tools such as Earned Value Management, Work Breakdown Structures, and Technical Performance Measures (Driessnack, 2017) offer valuable insight into the internal activities of acquisition organizations while helping PMs monitor and control program dynamics. However, these tools do not provide direct insight into the social and economic interactions that shape the organization's overall structure. To understand the bigger picture, PMs must rely on alternative information sources like organizational charts, employee interviews, and established institutional knowledge. Understanding the people, processes, and program perspectives that comprise an acquisition program is central to the core responsibilities of a PM and the overall success of the program; to that end, DoD has created and published guidance for PMs (DAU, 2022a, 2022b). While this guidance is applicable to most defense programs, its underlying assumption is that programs will follow the traditional structures outlined within the Defense Acquisition System (DAS).

Traditional program structures, such as those outlined within defense acquisition guidance, are hierarchical with predefined relationships and dependencies, making them relatively straightforward for the PM to navigate. When organizations cannot be easily defined by these hierarchical structures, as is the case with the DoD's software factories (Ryan, 2023), they can be referred to as nontraditional. The structures of nontraditional organizations can be difficult to navigate because their relationships and dependencies may not be as well-defined or documented, making it challenging for the PM to gain insights into the program's social and economic connections. This raises the question: If existing guidance is insufficient for nontraditional programs, how can the PM begin to understand the organization's structure and develop a strategic perspective? The more limited context of DoD software programs poses a more specific problem: *How can PMs assess the interorganizational structures of nontraditional software organizations?*

In this article, the aim is to address the challenge of understanding the structures of nontraditional programs by proposing and demonstrating a method of analysis that captures and visualizes interorganizational networks. Specifically, a methodology called Interorganizational Network-Analysis (ION-A) is presented, which utilizes multiplex, egocentric network analysis techniques¹ to capture the social and economic relationships of a program of interest.

Designing a network analysis study is a complex undertaking that requires the researcher to consider multiple attributes, data collection techniques, and measurement approaches prior to conducting an analysis (Provan et al., 2005). Combined with the fractured and difficult-to-navigate nature of network analysis (Popp et al., 2014), this can present a barrier to those seeking practical network analysis techniques to better understand their organizations. ION-A addresses this issue by defining the scope of the study, the types of networks captured, and the techniques of analysis to specifically assess nontraditional DoD software programs. Demonstrated through the analysis of the social and economic structures of a DoD software factory, this approach is intended to streamline the network analysis process and make it more accessible to PMs.

By applying the proposed ION-A process, the PM will be able to identify key relationships, dependencies, and channels of communication critical to the program's success, enabling better decision-making and resource allocation, and addressing potential issues arising from the program's unique structure. These factors are particularly relevant considering the DoD's emphasis on improving software acquisition processes and improving interorganizational trust and collaboration within its software factory ecosystem (DoD, 2023). The software factory, with its relatively flat organizational structure, is a prime candidate to demonstrate the utility of ION-A in assessing and understanding the nuances of a program's social and economic connectivity. The article concludes with a discussion of the research implications, a list of strategic analysis questions, areas for future research, and applicability of the ION-A approach to other nontraditional defense programs.



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Background

DoD software factories often adopt the network-based interorganizational structures commonly found in high-technology industries (Ryan, 2023). These structures influence the economic and social relationships among organizations differently than those defined by traditional

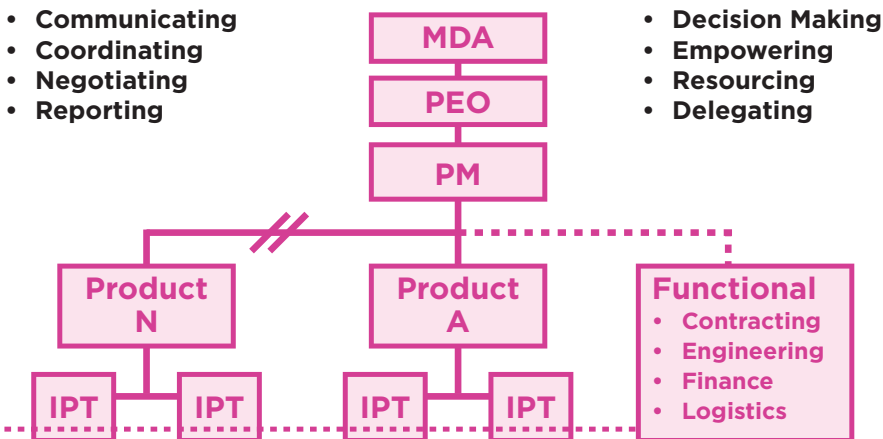
acquisition program boundaries. As a result, nontraditional programs do not always align with the established hierarchical structures outlined within the DAS. This case history analyzes one such program, a nontraditional Department of the Air Force (DAF) software factory, using a process called ION-A. This background section introduces the concept of interorganizational networks by explaining how they relate to familiar organizational models. It then describes their composition and explains how interorganizational networks can be used to visualize the economic and social relationships of an organization. Finally, this section closes by introducing the ION-A process as a method to characterize these relationships.

Visual models of interorganizational structures allow acquisition practitioners to easily draw on past experience and training to gain a baseline understanding of the behaviors and dependencies of a target organization. To illustrate this effect, we break down Figure 1, which depicts a common organizational structure composed of boxes and lines representing entities and their relationships. Because the structure of boxes and lines is easily recognizable and familiar to practitioners, the roles, motivations, and influences of the depicted entities can be assumed even if not explicitly stated on the diagram. Additionally, the basic economic and social interactions between entities can be inferred. Figure 1, in its simplicity, begins to frame the big picture of a program by invoking a familiar structure by which to understand economic and social relationships. However, when organizations do not follow these established patterns, the underlying assumptions regarding the economic and social relationships inferred from the hierarchical structure may no longer be valid. Therefore, additional steps should be taken to deliberately characterize the behaviors and patterns of exchange that were previously assumed or inferred.



FIGURE 1. ORGANIZATIONAL STRUCTURE

Organizational Structure—What resources do we need?



Note. DoD literature is primarily oriented to acquisition programs that follow formally defined hierarchical structures. In this chart, financial resourcing is implicitly tied to the PEO and MDA. Nontraditional programs like software factories can exhibit alternative structures that do not align with this framework. As a comparison, the PM of a traditional program may request additional funding through formal channels, while the PM of a software factory may primarily seek funding directly from customers or partner organizations. From *Program Manager (PM) Toolkit*, by J. Driessnack, 2017 (<https://www.dau.edu/tools/Lists/DAUTools/Attachments/143/Program%20Manager%20Toolkit.pdf>).

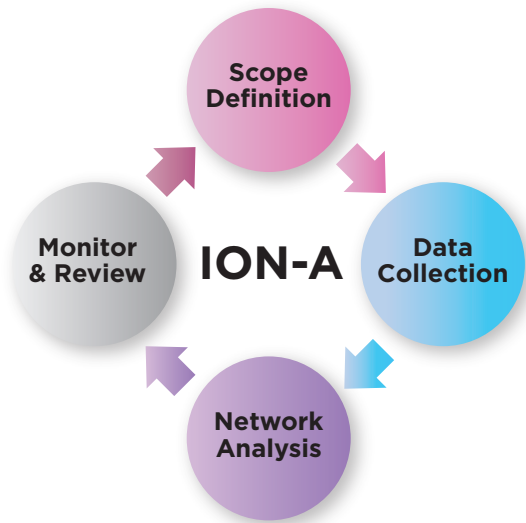
IPT = integrated product team; PEO = program executive officer; PM = program manager; MDA = milestone decision authority.

The generic program structure depicted in Figure 1 provides a visual representation of the primary interorganizational relationships of a program. If this representation was expanded to represent all organizations that interact with the program, it would begin to depict the program's network. By adding detail and defining the lines to represent relationships between the entities, we can formally define the resultant representation as the program's *interorganizational network*. An interorganizational network can be depicted as a collection of nodes, representing organizations or actors, and ties (also referred to as links or edges), representing relationships (Brass et al., 2004; Provan et al., 2005). These relationships, which can be observed through exchanges,² can be further categorized by type. Economic relationships can be observed by identifying exchanges of money and goods and services. Social relationships can be observed by identifying exchanges of information. When multiple types of relationships are represented within an interorganizational network, the network is defined as *multiplex*. When

viewed holistically, this collection of nodes and ties provides a model of the organization’s economic and social structures that can be analyzed.

Modeling and analyzing an organization’s interorganizational network can be accomplished by conducting an ION-A. This 4-step process, depicted in Figure 2, outlines how practitioners can deliberately capture and visualize the interorganizational networks of their programs; these networks can then be analyzed using established network methods and by employing a deliberate assessment approach designed to facilitate strategic thinking. The following sections further define the ION-A process and demonstrate its application in a case history analysis of a DAF software factory.

FIGURE 2. INTERORGANIZATIONAL NETWORK ANALYSIS



Note. This 4-step process can be used to develop a big picture view of an organization. ION-A focuses on understanding the social and economic relationships of a target program by visualizing exchanges of money, goods and services, and information.

Situational Awareness Approach

To demonstrate how the ION-A process can be utilized to better understand nontraditional software programs, we embedded ourselves within a DAF software factory that could not be easily characterized via the formally established structures outlined within the DAS. To prepare for this case history, develop a contextual understanding of the organization’s patterns of behavior, and establish trust, we observed the daily stand-up meetings of the software factory leaders for 3 months. This approach simulates the experience of someone new to the organization and suggests

that practitioners do not need extensive experience within the program to conduct a meaningful analysis.

The software factory analyzed for this case history has existed for 1 year; it has no formally defined military or civilian manpower requirements. Instead, this organization consists of participants from partnering organizations who have volunteered to manage and grow the software factory. Its purpose is to provide a specific niche of goods and services in the form of consulting services, design services, educational outreach, application development, and systems development and maintenance to external customers.

Since the software factory is in the early stages of growth, it relies heavily on organizations within its local network to supplement core functions such as contracting, finance, and legal services. This organization also maintains autonomy and control over its assets and has been formally recognized by the DAF Chief Software Office (CSO) as a software factory. Programmatically, this software factory operates outside of existing acquisition pathways and does not have the compulsory reporting requirements of a formal program.



Many of the strategies employed by the software factory of interest are in direct alignment with the principles outlined within DoDI 5000.87, Operation of the Software Acquisition Pathway (DoD, 2020, pp. 11–12). The software factory's use of existing enterprise services and contracts, its employment of a modular contracting strategy, and its stakeholder-centric alignment are all considered key elements of a modern software program operating within the new software acquisition pathway. Collecting information on software factories was challenging because many candidates were uneasy about sharing organizational information. We addressed this concern by nonattribution.

Step 1—Scope Definition

Step 1 in the ION-A process is to define the scope of analysis, which begins by defining the nodes and ties of the interorganizational network. Nodes will exist in all organizations that interact with the program, and ties will represent relationships between those organizations. When defining nodes, the specific boundaries of the individual military organizations they represent will need to be considered based on the size and environmental context of the program of interest. To maintain a strategic focus in this analysis, military organizations were aggregated and represented as nodes at their respective branch or squadron levels. Three types of ties, representing both economic and social relationships, are defined using the exchange variables: goods and services, financial resources, and information. A detailed definition of these variables is included in Table 1.

TABLE 1. NETWORK VARIABLES OF EXCHANGE		
Exchange Variable	Definition	Relationship Type
Goods & Services	<i>The transfer of goods & services from one organization to another.</i>	<i>Economic</i>
Financial	<i>The transfer of money from one organization to another.</i>	<i>Economic</i>
Information	<i>The active and deliberate transfer of meaningful information, knowledge, communication, or coordination where one or both parties benefit.</i>	<i>Social</i>

Note. The variables (goods and services, financial, and information) were selected based on the findings of recent research into the structures of software factories (Ryan et al., 2022) and research on multiplex economic networks (Maghssudipour et al., 2020).

Step 2—Data Collection

The second step in the ION-A process involves collecting and organizing data. Data collection presents a significant challenge in the ION-A process, but it can be divided into two primary tasks: identifying nodes and identifying ties. Node identification starts with generating a list of all known organizations that have interacted with the program. This initial list can be compiled from various sources, such as internal websites, existing documentation, or employee interviews. In the analysis of the software factory, the list was created by identifying organizations mentioned within Confluence, the software factory’s internal knowledge management system. Each organization was categorized by affiliation as government, industry, or working group, and organized in a matrix format using Excel. Figure 2 illustrates the creation, organization, and interpretation of a matrix.

Upon creating a matrix, the next step is to identify interorganizational exchanges (Figure 3). Since a consolidated data source containing a comprehensive listing of exchanges is unlikely to exist, data collection should

be conducted through semistructured interviews or brainstorming sessions with experienced personnel, using the matrix as a reference.

FIGURE 3. A SOCIOMETRIC MATRIX

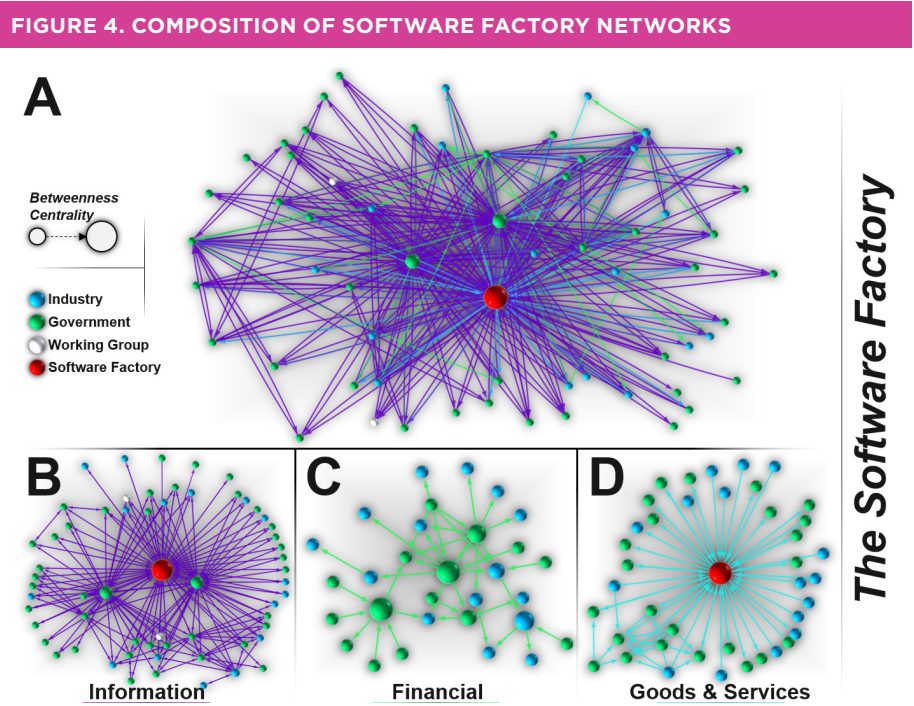
	Software Factory	Organization A	Organization B	Organization C	Organization D	Organization E
Software Factory				I	I, S	I, S
Organization A	S					
Organization B	S					
Organization C	I					I
Organization D	I					I
Organization E	I, S		I	I		
Organization F	I, S					I
Organization G	I					I, F
Organization H	I					
Organization I	I					
Organization J	I					

Note. The Sociometric Matrix can be used to capture exchanges between organizations. Organizations identified in the roster are organized in the first column and duplicated in the first row. Exchanges between these organizations are then noted within each corresponding cell of the matrix. The matrix is interpreted by reading “column A sends (money, information, goods and services) to Row 1.” Example: The software factory sends information and services to organization D.

In this case study, a series of video interviews were conducted with software factory leadership to identify interorganizational exchanges. The matrix served as a guide and tool for documenting individual exchanges. The interviewees were asked to provide information on exchanges between organizations at each intersection within the matrix. All exchanges were evaluated for directionality. The matrix was completed after four interviews, and responses were validated by reviewing the video recordings. Inconsistent or unclear responses were addressed in subsequent interviews. The final matrix consisted of 62 organizations and their exchanges.

Once the data are collected, they must be converted to a network format for analysis. Various existing open-source network tools can be employed for this task. In this case study, the finalized matrix was processed using the open-source network graphing software *Gephi* (Bastian et al., 2009) and *Cytoscope* (Shannon et al., 2003). Step 2 is considered complete once the data have been initially processed using network graphing software. The

networks of the software factory are depicted in Figure 4. The composite network (4A) is generated by integrating the three variables of exchange: information (4B), financial resources (4C), and goods and services (4D) into a single model.



Note. Four separate interorganizational networks were generated from the exchange variables, goods and services, financial, and information, collected in Step 2 of the ION-A. The software factory's composite network consists of government (69.3%), industry (25.8%), and working groups (3.2%), totaling 62 organizations. These networks provide a visualization of the economic and social structures of the program, which are analyzed in step 3 of the ION-A process.

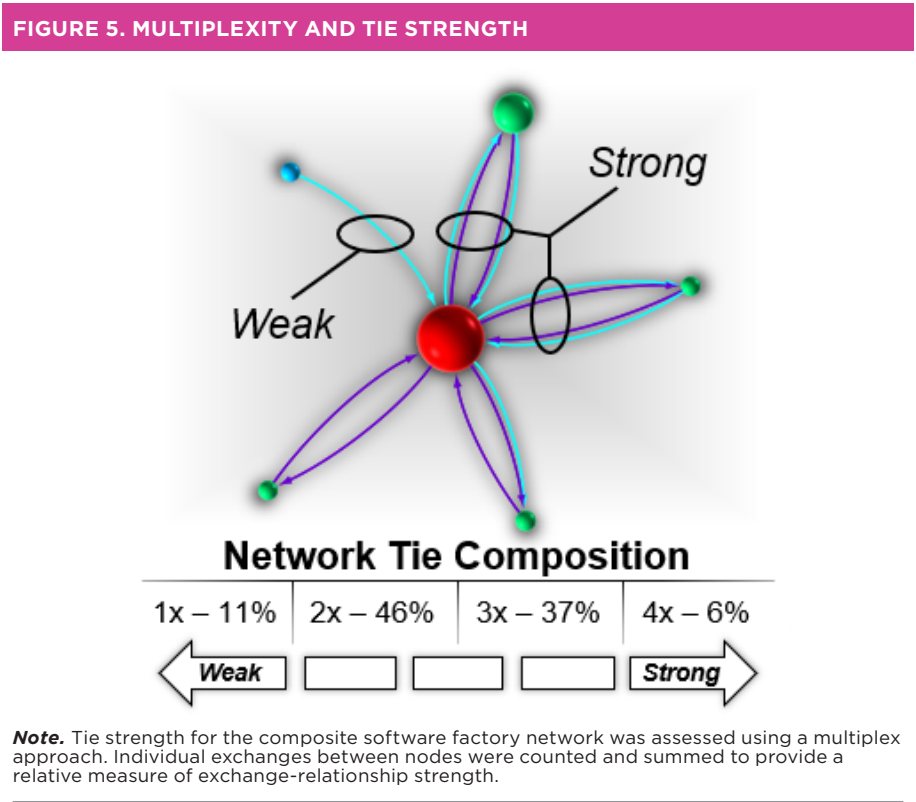
Step 3—Network Analysis

The third step in the ION-A process involves analyzing the inter-organizational network of the software factory. This stage focuses on evaluating four strategic areas: relationship strength, organizational influence, exchange patterns, and communities of interest. Table 3, presented in the conclusion, provides a summarized assessment framework that encompasses these four areas of interest along with assessment questions and common network indicators. The following discussion addresses these four areas.

Relationship Strength

The analysis begins at the composite network level (4A) by evaluating the strength of relationships among organizations using the concept of

multiplexity. The existence of multiple types of ties (i.e., social and economic) between entities is generally believed to indicate greater strength and durability of the relationship (Granovetter, 1973; Perry et al., 2018). By counting the ties between nodes, a basic assessment of strength can be determined. More ties suggest a stronger relationship, while fewer ties indicate a weaker relationship (Figure 5). Critical relationships can be identified manually or with network analysis software for a comprehensive assessment of the entire network.



The software factory’s network predominantly consists of ties with medium relative strength, with only 6% of ties reaching the strongest threshold. These ties, representing shared, reciprocal relationships, mainly exist between the software factory and its partnering organizations. Notably, this tie pattern is also present between the sample software factory and a government platform provider where a two-way exchange of services takes place. The weakest ties within the software factory network are primarily informational, with unidirectional reporting relationships serving as examples of weak ties.

Evaluating relationship strength offers valuable insights into common organizational interactions. By engaging in different types of exchange, strong ties can be intentionally established. As the organization matures, weak links can be identified and deliberately strengthened. Strategic social partnerships can be reinforced through economic exchange. For instance, the software factory provides software services at no cost to an organization controlling access to informational resources, which strengthens the software factory's relationship with an influential organization. This leads to the next area of analysis: organizational influence.



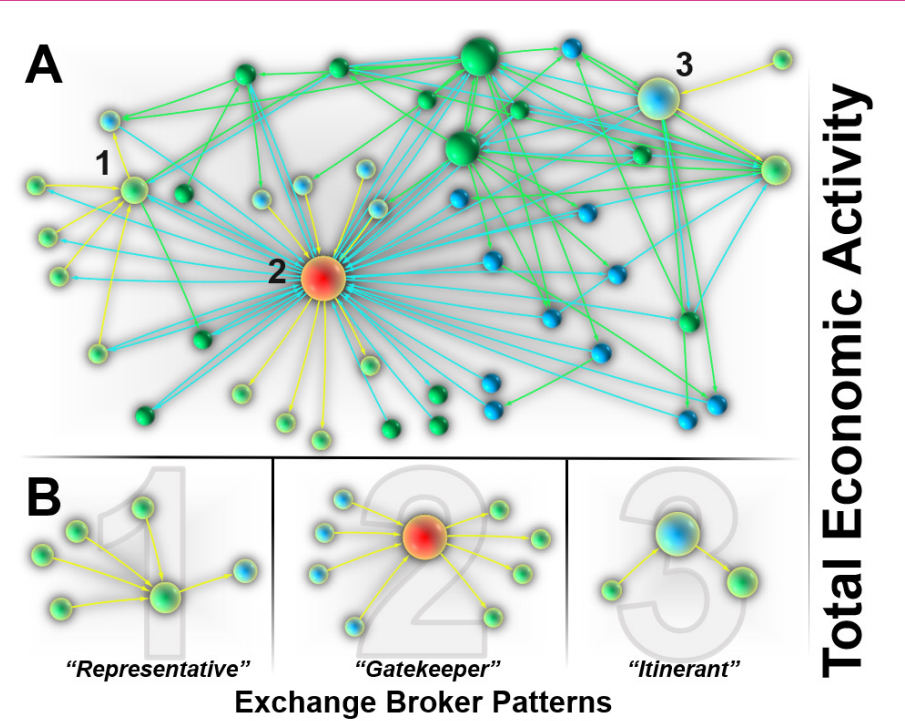
One notable observation is that the flow of financial resources through the network is varied; the software factory is funded by customer organizations rather than by a centralized source.

Organizational Influence

Most practitioners are familiar with the authoritative power structures that exist within a traditional hierarchy. However, in situations where these structures are less well-defined, an organization's positional power can often play a more significant role in its ability to exert power and control over the program. Positional power structures can be identified by analyzing centrally positioned organizations, critical resource flows, and exchange brokers within the program's interorganizational network. By considering these factors in conjunction with the program's requirements, it is possible to identify influential organizations.

To identify centrally positioned organizations, the composite interorganizational network of the software factory was assessed for betweenness centrality, which is represented on the network models using node size. Betweenness centrality measures how often a node exists on the shortest path between other nodes within the network, aiding in identifying nodes that have influence over the flow of information within the network due to their position (Freeman, 1977). While different types of centrality measures can be used, betweenness centrality was chosen because of its applicability to both sociocentric and egocentric network data (Marsden, 2002). The two primary partner organizations of the software factory had the highest centrality, indicating their importance relative to other organizations in the network. When assessing the economic network for centrality (displayed in Figure 6A), additional influential organizations can be identified.

FIGURE 6. ECONOMIC NETWORK OF THE SOFTWARE FACTORY



Note. This economic network is generated by combining the exchange variables financial and goods and services. In addition to identifying centrally positioned organizations, three broker roles were deemed notable within the software factory economic network: the representative, the gatekeeper, and the itinerant. Tracking these brokers is an essential component of identifying and controlling external program risks.

The economic network generated from the financial resources and goods and services exchange variables provides valuable insight into the flow of resources between organizations. One notable observation is that the flow of financial resources through the network is varied; the software factory is funded by customer organizations rather than by a centralized source. To utilize these resources, the software factory relies on multiple contract vehicles owned by other organizations within its network. The organizations that manage and maintain these vehicles function as economic hubs. Because they control the flow of resources from customers to the software factory, they have positional power. Another observation on the flow of goods and services is largely centered around the software factory, with additional convergence occurring around the organizations acting as brokers, which are organizations that help coordinate exchange.

The software factory network (highlighted in Figure 6B) embodies three primary broker roles: the representative, the gatekeeper, and the itinerant. These roles are defined based on their affiliation and positioning within the network (Gould et al., 1989). The *representative* broker emerges when a government organization coordinates a transaction from another government organization to an industry partner. In this scenario, one government organization *represents* the other. This situation occurs when customers fund a contractor through a third-party government organization. The *gatekeeper* role exists when a government organization purchases services from a third party via a contractor organization. The contractor functions as a *gatekeeper* between the third party and the purchaser. Gatekeeping also occurs when an industry partner transfers services to a government organization, which then passes the services to a government customer. The software factory primarily functions as a gatekeeper. The final brokerage role, the *itinerant*, arises when an industry partner acts between two government organizations. All itinerant brokerage relationships within the software factory network are managed through formal contract vehicles. The influence of brokers must be considered in conjunction with their organizational affiliation and the flow of resources through the network.

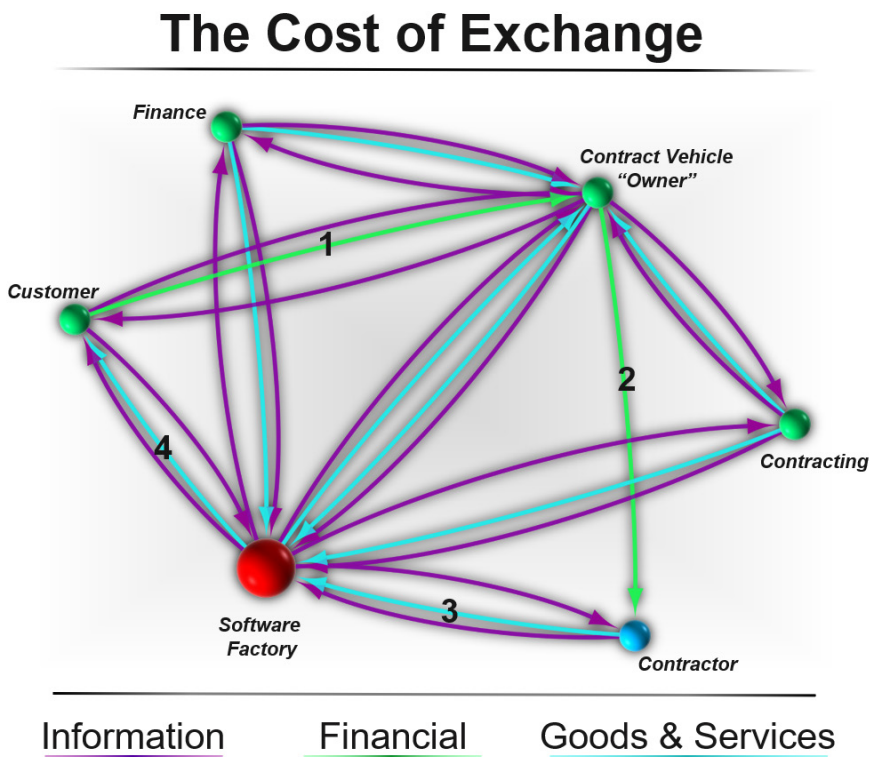
In situations where traditional hierarchical structures are not clearly defined, identifying the organizations that hold the most influence over a program can be challenging. However, ION-A can aid practitioners in this task by providing a visual representation of the program's interorganizational relationships. By analyzing these resulting models, researchers can assess centrality, identify resource flows, and determine broker roles. This information can then be used to pinpoint influential organizations. Next, the analysis will explore how identifying patterns of exchange can inform the big-picture perspective of a program and assist in decision-making.

Exchange Patterns

At this point in the ION-A process, significant organizations and their relationships within the program's network have been identified and assessed with the help of network-based measures such as multiplexity and centrality. In contrast, identifying exchange patterns demands a more refined analytical approach. Instead of relying on specific measures, it is important to identify exchange patterns needed to complete known inter-organizational transactions. Additionally, since the relative importance of specific transactions is unique within each program, the program's mission statement, its organizational objectives, and its core functions should also be considered. The analysis in this paper concentrated on the transactions between the software factory and its customers due to their frequency and relative importance to the program's strategy.

Figure 7 illustrates the exchanges required to execute a typical transaction between a customer organization and the software factory. In addition to the primary exchanges, secondary exchanges are arising from support functions such as contracting, finance, and legal services associated with the execution of funds. Since the software factory does not maintain these functions organically, it engages external organizations' help. These exchanges, which must be coordinated by factory personnel, represent transaction costs, which are sometimes neglected using internal government analysis methodologies.

FIGURE 7. EXCHANGE FLOW AND TRANSACTION COSTS



Note. Four primary exchanges must occur to complete a transaction between the software factory and a customer: (1) the customer sends money to a third-party organization that maintains an existing contract vehicle; (2) money is sent to the contractor via the vehicle; (3) the contractor provides services to the software factory; and (4) the software factory provides services to the customer organization. In summary, the software factory orchestrates every step of the transaction through its social and economic connections.

After analyzing the transactions of interest, network models should be assessed holistically to identify areas of activity that seem unusual or out of place. When the reasoning behind an exchange pattern is unknown,

experienced members of the organization should be consulted and asked clarifying questions using the network model as a visual guide. Finally, an evaluation of exchange patterns should be conducted to determine if any can be simplified to reduce the overall cost of individual transactions. Step 3 of the ION-A process concludes with an assessment of the communities within the program's interorganizational network.



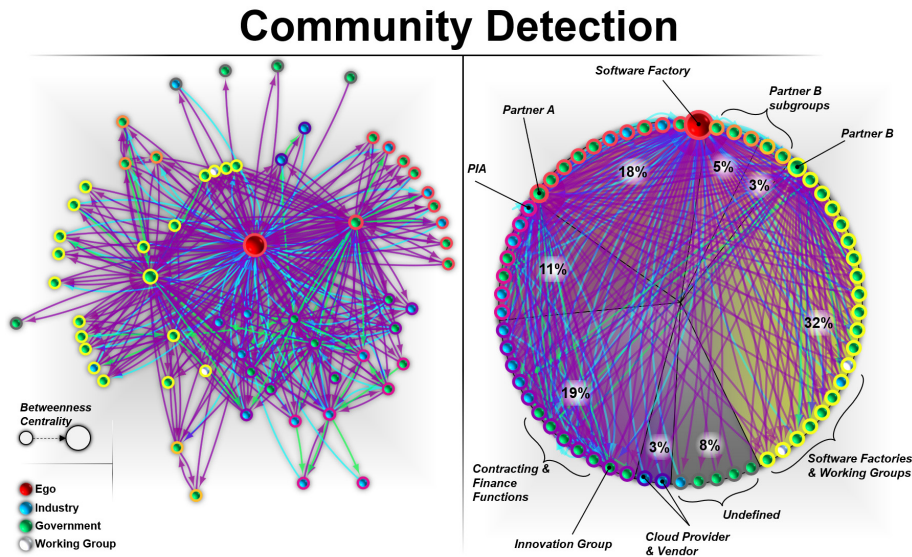
Examining exchange patterns can help identify dependencies on economic or social resources, enabling the development of proactive mitigation strategies in line with risk management best practices.

Communities of Interest

Identifying communities within the program's interorganizational network allows practitioners to comprehend and navigate the intricate relationships and interdependencies that exist between organizational subgroups. Community detection algorithms can quickly identify these communities. A hierarchical community detection algorithm (Pons & Latapy, 2005) was applied to the software factory network; seven distinct communities of interest were highlighted (Figure 8). The largest communities roughly aligned with the parent commands of the two partner organizations, activity associated with a local "innovation group," and a community of organizations associated with an industry partner acting under a Partnership Intermediary Agreement (PIA). The algorithm also highlighted three additional smaller communities: two of these communities can loosely be categorized as subgroups of Partner B; one highlights a gatekeeper brokerage pattern between an industry partner and a cloud vendor.



FIGURE 8. COMMUNITY DETECTION



Note. A random walk community detection algorithm was utilized to identify communities of interest within the composite software factory network. Community detection algorithms identify subgroups within a community by measuring connections between nodes. When applied to the software factory network, seven separate clusters were identified. While many clusters were easily attributable to known communities, others provided additional insight by highlighting previously unknown subgroups.

When applied to the software factory network, the community detection algorithm accurately identified known communities, indicating its validity as an assessment tool. Additionally, its identification of smaller subgroups within the software factory network provided an alternative perspective on subgroup dynamics. By identifying communities of interest early in the program life cycle, the PM can plan the organization's growth to align with community needs.

The analysis step of the ION-A process is now complete. This step utilized the network models created in Step 2 to identify influential organizations, organizational relationships, common exchange patterns, and communities of interest within a nontraditional program's interorganizational network. This information can provide the PM with a foundational understanding of the interorganizational dynamics of their program and can then be used to establish an informed perspective.

Step 4—Monitor and Review

The fourth step in the ION-A process involves ongoing monitoring and assessment of the interorganizational network. PMs should regularly update

the network analysis to accommodate new contextual information, as networks are dynamic and subject to change over time. By frequently updating the network data, managers can obtain a measurable visualization of organizational evolution and growth.

This section demonstrates how practitioners can evaluate a program's inter-organizational network to gain insights into organizational interactions and behaviors. Identifying influential organizations and relationships can inform stakeholder management plans and aid in prioritizing future interactions. Examining exchange patterns can help identify dependencies on economic or social resources, enabling the development of proactive mitigation strategies in line with risk management best practices. Community identification can offer insights into social dynamics and uncover previously unknown communities of interest.



In this specific case history, the results of the network analysis process (summarized in Table 2) were used to establish a shared internal understanding of the interorganizational relationships and behaviors of the program. The network graphs created during the analysis were also used to help communicate the intangible scope of software factory operations and program interdependencies to senior leaders outside the organization. Additionally, the software factory was able to use the analysis to identify enterprise-level, high-cost interorganizational processes and communicate how these costs could be reduced by increasing internal resources.

TABLE 2. THE ION-A PROCESS

Influential Organizations	Relational Assessment	Exchange Patterns	Community Assessment
Partner Organizations A & B, Contract Vehicle Owners, Financial Brokers, Cloud Contract Liaison, PIA Org, Innovation Group	Weakest Relations: Unidirectional information reporting, Brokered vendor relations Strongest Relations: Factory Partners, Factory—Platform Provider, Factory—Personnel Services Org	Customer—Factory Transaction: Requires 6 Orgs & minimum 4 economic exchanges; 3 industry partners funded through multiple sources; financial convergence through contract vehicle owners	Four dominant communities: Partner A, Partner B, Innovation Group, & PIA; 3 subgroups: 2 Partner B & Cloud Vendor

Strategic Insights:

- Industry partners are funded through multiple sources/customers with competing expectations, interests and requirements: this complicates program-level cost, schedule, and performance metrics/assessments. Alternative, customer-centric metrics should be developed.
- The software factory primarily acts in the "gatekeeper" broker role. Industry partner services are rendered to customers through the software factory: this is an unstated benefit/service of the software factory structure. Consider emphasizing this role when engaging potential customers.
- The software factory relies on many organizations within its network for resources (financial, services, and information): this prevents power centralization and provides increased autonomy to the factory at a cost to organizational stability.
- Accounting data are fragmented and dispersed across five organizations and two MAJCOMs: this makes it difficult to communicate total program cost.
- The software factory relies heavily on its social network to operate: this places a strain on key members of the organization (Cook, 1977) and limits organizational mobility (Blau, 2017). Consider implementing a customer liaison/outreach role to manage social exchanges.
- The social costs required to execute economic exchange are high: the factory should identify ways to reduce the cost of customer-factory transactions. Transaction steps are currently "tribal" knowledge and they should be formally documented to combat knowledge attrition.
- The relationships between cloud providers are weak and rely on third-party industry partners: this introduces a risk to a critical resource.
- Partner communities dominate the interorganizational network: the software factory should strengthen relationships with additional communities to reduce partner dependencies.

Note. This process provides practitioners with a method to assess the social and economic relationships of their programs. This case history assessed an existing DAF software factory in four key areas: organizational influence, relationship strength, exchange patterns, and communities of interest. Strategic insights, organizational risks, and potential opportunities tailored towards the specific needs of this nontraditional software factory organization are derived from the ION-A process.

MAJCOMs = major commands; PIA = Partnership Intermediary Agreement.

In summary, the ION-A process provides acquisition practitioners with a strategic perspective on the social and economic relationships within their programs. This section illustrated this point by presenting a case history of a nontraditional software program showcasing the practical benefits of network analysis. Table 2 presents a summary of the most relevant outputs and strategic insights gained from the ION-A process.

Limitations and Future Work

The aim of this case study was to demonstrate a practical network analysis process capable of capturing and understanding the social and economic relationships of nontraditional software programs. This process, termed ION-A, streamlines the network analysis process by predefining the scope of the analysis, the variables of interest, and the analysis techniques employed, reducing the initial research and planning requirements for PMs. While this predefinition makes network analysis more accessible, it also limits the applicability of ION-A to cases similar to the demonstrated case. Furthermore, the field of network analysis provides countless techniques, tools, and analysis methods available to researchers, which have not been discussed. For instance, alternative measures of relationship strength (e.g., dollars exchanged, frequency of exchanges) and organizational influence (e.g., eigenvalue centrality, closeness) could be employed. These and many other techniques present an exciting opportunity for future research as they may reveal additional insights into the organizational dynamics of programs.

Conclusions drawn regarding the specific software factory analyzed within this case study also face limitations. First, the results generated from this network analysis are highly temporal. Organizations are constantly evolving, and this analysis represents a single snapshot in time. Second, the results from this network analysis cannot be used to assess the validity or effectiveness of the nontraditional organizational structures employed by the DoD's software factories. Although this study provides insight into the relationships of the featured organization, it only captures a single data point in the context of the broader software ecosystem. Future studies could utilize the framework established within this study to capture and analyze the ego networks of multiple organizations within the ecosystem. This would allow for a more in-depth statistical analysis of the programs' interorganizational networks.

Lastly, this study is constrained by data availability. Data on software factory performance are limited. As a result, the data employed in this study were gathered through interviews with members of the software factory of interest, and the members' bias can be assumed based on their perspectives, organizational goals, and opinions. The analyzed organization is in its infancy; inevitably, it is in the interviewees' best interest to present a positive view of their organization. Although this study implemented a structured approach to mitigate this effect, it cannot be eliminated entirely. Due to these limitations, the analysis process presented in this case study

can only be used to identify the existence of organizational characteristics and behaviors for the organization under examination.

Future research could explore the integration of additional network analysis techniques and measures of relationship strength (e.g., dollars exchanged, frequency of exchanges) or organizational influence (e.g., eigenvalue centrality, closeness) to further refine the ION-A methodology. Additionally, researchers could investigate the applicability of ION-A to a broader range of nontraditional defense programs like those operating under the software acquisition pathway or compare the effectiveness of ION-A to other existing network analysis approaches. Through ongoing research and development, the ION-A methodology can continue to evolve and provide program managers with increasingly effective tools for assessing and understanding the social and economic connectivity of their programs.

Conclusions

This article began by highlighting that existing guidance for acquisition programs is tailored toward organizations with traditional hierarchical structures; thus, it may not be sufficient for assessing nontraditional programs. This led to the formulation of a research question designed to address this gap in understanding: *How can program managers assess the interorganizational structures of nontraditional software organizations?*

To answer the research question, we developed and applied a process called ION-A and demonstrated its effectiveness using the case history method on a nontraditional DAF software factory. The ION-A process, which employs the network analysis principles of multiplexity and egocentricity to capture the economic and social relationships of a single program of interest, has been simplified for practicality yet still provides a technically sound foundation for practitioners to analyze and understand the organizational structures of their programs.

The ION-A process describes how practitioners can collect and organize network-centric program data that can be used to model the social and economic relationships of their programs. It also describes data collection techniques and identifies open-source software programs that can be used to create an interorganizational network from the collected data. The ION-A process progresses by guiding practitioners through an analysis of four focus areas of interorganizational networks using a framework of contextual assessments. Table 3 includes a summary of this framework, which addresses these four focus areas, provides initial assessment questions, and identifies a collection of relevant analysis techniques and network models that can be used in the analysis.

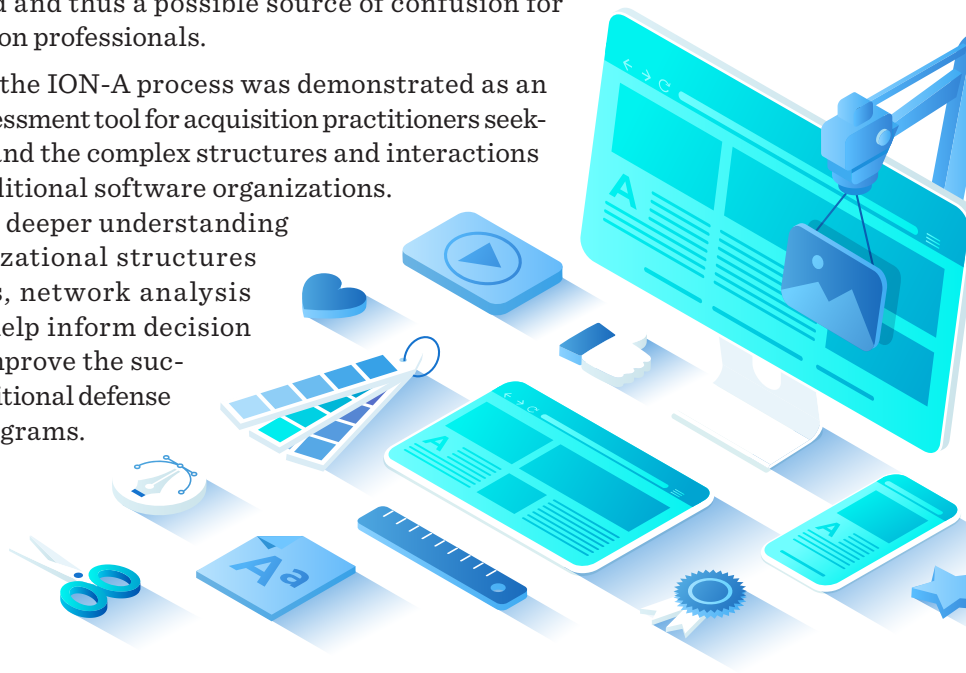
TABLE 3. THE ION-A ASSESSMENT		
Area of Interest	Contextual Assessment	Network Indicators
Relationship	Identify the weakest and strongest relationships within the organization's network. a. Are the social and economic costs of these relationships in alignment with the goals of the organization? b. Can strategic relationships be strengthened by establishing additional types of exchange relations? Should they? c. Are any interorganizational relationships stronger or weaker than anticipated? Why?	Composite Network (Figure 4A), Multiplexity & Tie Strength (Figure 5)
Organizational Influence	Identify the most influential organizations within the network. Consider both the social and economic environments. a. Which organizations are centrally located? Why? b. Which organizations are acting in broker roles? What is their affiliation? c. Which organizations control access to resources? Information? Is this an acceptable risk? d. Does the flow of resources converge at any specific organization? e. What happens if these organizations exit the network?	Economic, Social, and Composite Networks (Figure 4A-D), Centrality Measures, Network Positioning, Resource Flows, Broker Roles (Figure 6)
Exchange Patterns	Identify common patterns of exchange. a. How many exchanges must occur to complete a transaction? Should this be reduced? b. Are critical patterns of exchange formally documented? c. Are there any patterns of exchange that could be standardized? Simplified? d. Are economic exchanges supported by social exchanges? If not, why?	Economic, social, and composite networks (Figures 4A-D), Exchange patterns (Figure 7), Resource flows, Directionality
Communities of Interest	Identify clusters of organizations or communities of interest. a. Is it possible to explain the existence of each community? b. Are there any subgroups that were previously unknown? c. Do any organizations seem misplaced? Why? d. Are communities clustered around specific organizations? What are they?	Composite Network (Figure 4A), Community Detection (Figure 8)

Note. This process guides practitioners through an assessment of the interorganizational network of their programs. The analysis framework used in Step 3 of ION-A focuses on four primary areas of interest in order to provide a foundational understanding of program dynamics. Questions designed to guide the assessment are provided along with relevant network-based indicators.

In addition to developing and demonstrating a new assessment approach, this case history also provided a network-centric look into the underlying mechanisms driving nontraditional software factories. While these insights hold value, performing a network analysis can be time consuming. Nonetheless, advancements in network management continue to increase their worth, resulting in practical applications for industry practitioners to gain fresh insights into organizational behaviors (Bodan et al., 2016; Cross & Gray, 2021). Open-source graphing applications, which aided the analysis in this case history, also continue to evolve and become more accessible. As a result, practitioners' ability to apply network analysis techniques, such as ION-A, to gain insights into their organizations will keep growing. Cultivating and managing interorganizational partnerships and stakeholder relationships is crucial for any program and network assessment techniques like those described in this article, which enable practitioners to better understand these relationships early in the program life cycle.

This article contributes to both the Department of Defense and academia by demonstrating a structured approach to organizational analysis while also providing an in-depth look into an emergent acquisition organization, the software factory. This case history also provided insight into the structure and relationships of an organization prior to a formal entry into the acquisition system. Identifying alternative analysis methods for such organizations is important since this early period is largely undocumented and thus a possible source of confusion for many acquisition professionals.

In conclusion, the ION-A process was demonstrated as an alternative assessment tool for acquisition practitioners seeking to understand the complex structures and interactions within nontraditional software organizations. By providing a deeper understanding of interorganizational structures and behaviors, network analysis methods can help inform decision making and improve the success of nontraditional defense acquisition programs.



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Endnotes

¹ Multiplex, Egocentric Network Analysis: A multiplex network displays multiple types of relationships or ties between organizations. Egocentric networks focus on a specific organization of interest, defined as the ego (Perry et al., 2018). This differs from a sociocentric analysis approach, which seeks to derive insights into the whole network. Multiplexity is discussed in depth in the analysis section of this article.

² The utilization of the exchange as a way to measure organizational relations was initially proposed as a method for studying interorganizational behavior and relations in the early 60s by Levine and White (1961). Since its initial definition, the exchange has been broadly employed by researchers to measure both social and economic transactions between both individuals and organizations (Wasserman & Faust, 1994).

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